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DATA SCIENCE AI350

U.S. Domestic Flight Delay Analysis: A Data Quality and Performance Study Using BTS On-Time Reporting Data (2025)

Research Proposal | The data miners | March 2025

1. Research Question & Objectives

The objectives of the present research are as follows: "What are the principal causes and trends of flight delays for U.S. domestic air carriers, and how might improvements to data quality reduce the uncertainties associated with the analysis of on-time performance using BTS reporting data?"

Key objectives:

- Identify and resolve data quality issues including missing values, outliers, duplicates, and inconsistencies.
- Analyze delay patterns across airlines, airports, routes, and seasons.
- Benchmark on-time performance across 14 U.S. domestic carriers.
- Develop and evaluate a machine learning model to predict flight delay likelihood.

2. Dataset

The dataset was obtained from the **Bureau of Transportation Statistics (BTS)**, a federal agency under the U.S. Department of Transportation (transtats.bts.gov), sourced from the *Reporting Carrier On-Time Performance* database — which mandates monthly flight-level reporting from all certified U.S. carriers.

The three monthly extracts, January, June, and November 2025, represent the winter, summer, and autumn seasons. This provides a combined 1,650,044 structured flight records, 32 attributes, and a CSV file. A stratified random sample of 300,000 records, 100,000 from each month, will be extracted to ensure the integrity of the data distribution and computational efficiency.

Redundant features, such as YEAR, FLIGHTS, DEP_DELAY_GROUP, and ARR_DELAY_GROUP, will be removed. Derived features, Airline_Name, Season, is_Delayed, Route, and Scheduled_Hour, will be engineered for greater analytical value.

3. Methodology

Tools: Python, Pandas, NumPy, Scikit-learn, Matplotlib/Seaborn/Plotly, colab Notebook.

- Phase 1 — Data Integration: Merge 3 CSV files, apply stratified sampling, derive Airline_Name mapping from IATA codes.
- Phase 2 — Data Cleaning: Treat missing values (mean/median/KNN imputation), detect outliers (IQR, Z-Score, Isolation Forest), remove duplicates (exact & fuzzy matching), fix logic errors (e.g. cancelled flights with departure delay).
- Phase 3 — EDA: Distribution analysis, delay heatmaps by airline/season, route-level rankings, cancellation pattern analysis.
- Phase 4 — Modeling: Binary classification (Is_Delayed: DEP_DELAY > 15 min). Three models compared:

Model	Role	Key Strength
Logistic Regression	Baseline	Fast & interpretable
Random Forest	Intermediate	Accuracy + feature importance
XGBoost	Final	Best tabular performance

Evaluation: Accuracy, Precision, Recall, F1-Score, ROC-AUC, Confusion Matrix. Class imbalance handled via SMOTE. 5-fold cross-validation applied.

4. Expected Outcomes

- A fully cleaned validated dataset free of quality issues — ready for future aviation research.
- Identification of the most significant delay contributors across airlines, airports, and seasons.
- A comparative performance ranking of all 14 U.S. carriers and major airports.
- A validated predictive model with documented accuracy metrics and model comparison.
- A replicable data quality pipeline contributing to best practices in applied data science.

5. References

[1] Bureau of Transportation Statistics. On-Time Reporting Carrier On-Time Performance. DOT.
<https://transtats.bts.gov>

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